Which Backbone Picks Which Wordification? A Factorial Study of Topic-Model Families on Hyperspectral Imagery

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***Abstract*—The companion paper showed that twelve wordification recipes V1–V12, plus a learned VQ-VAE recipe V13, produce measurably different topic bases under a fixed LDA backbone — V12 (Gaussian-mixture tokens) wins on classification, coherence, label coupling and counterfactual robustness, while V1 (band-frequency tokens, the canonical) wins on seed-test reliability. The present manuscript asks the orthogonal question: *does the recipe winner depend on the topic-model backbone?* We run the same V1–V13 sweep under three additional backbones — Hierarchical Dirichlet Process (HDP, nonparametric K), Product LDA (ProdLDA, logistic-normal prior + amortised inference) and the Embedded Topic Model (ETM, joint word–topic embedding) — producing a 4×13 factorial on six labelled-scene benchmarks (312 cells in the headline F-2 coherence axis). The headline finding is that the wordification ranking inverts across backbones. LDA and ETM, both with Dirichlet-like priors, agree that V12 wins; HDP with its stick-breaking truncation prior picks V7 (absorption-feature triplets); ProdLDA with its logistic-normal prior picks V3 (joint (band, bin) vocabulary). We give a per-recipe *affinity score* that quantifies how each recipe transfers across backbones, identify three robust backbone–recipe pairings, and recommend reporting the $(\text{backbone}, \text{wordification})$ pair as a single hyper-decision in any future LDA-on-HSI work. LDVAE-T is the fifth proposed backbone but is parked pending public code.**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, ProdLDA, ETM, HDP, wordification, topic models, factorial study, evaluation framework, backbone choice, learned tokenisation.**

I. Introduction

The first two manuscripts in this programme (P1, P3) hold the topic model backbone fixed (online-VB LDA) while varying the *wordification recipe* that converts a pixel spectrum into a multiset of LDA-consumable tokens. P3 shows that the recipe matters on every F-axis of the twelve-axis evaluation framework introduced in P1, with a single recipe (V12, Gaussian-mixture-component tokens) winning four of six F-axes under LDA.

This paper inverts the question. We hold the recipe *set* $\{V1, \ldots, V13\}$ constant and vary the backbone across four topic-model families: LDA (baseline carried over from P3), Hierarchical Dirichlet Process (HDP, Teh et al. 2006), Product LDA (ProdLDA, Srivastava & Sutton 2017), and the Embedded Topic Model (ETM, Dieng et al. 2020). All four share the basic generative structure — topics are distributions over a discrete vocabulary, documents are mixtures of topics — but each carries a different prior over the topic simplex, a different inference procedure, and in ETM's case a different way of parameterising the topic–word distribution as a softmax of embedding dot products.

A. The headline finding

The wordification ranking is *not* invariant under the backbone choice. Table I summarises the per-backbone F-2 coherence winner across the six labelled-scene panel. LDA and ETM agree on V12; HDP picks V7; ProdLDA picks V3. Three of four backbones disagree on the single best recipe.

TABLE I
PER-BACKBONE F-2 c_v WINNER ON THE SIX LABELLED-SCENE PANEL (UNIFORM / $Q = 8$, MEAN ACROSS SCENES).

Backbone	Best recipe	Mean c_v
LDA	V12 (GMM-token)	0.85
HDP	V7 (absorption triplet)	0.615
ProdLDA	V3 (joint band–bin)	0.863
ETM	V12 (GMM-token)	0.816

This is the formal version of the observation that the wordification choice and the backbone choice cannot be made independently. P3's recommendation that V1 retain canonical-default status assumed an LDA backbone; the present manuscript shows that switching backbone changes the practical canonical too.

B. Contributions

1) A 4×13 factorial sweep over four topic-model backbones (LDA, HDP, ProdLDA, ETM) and thirteen wordification recipes (V1–V13) on six labelled HSI scenes. 312 F-2 cells, 312 F-14 cells.
2) Identification of three robust backbone–recipe affinities: Dirichlet-prior backbones favour dense-vocabulary recipes (V12, V3); truncation-prior backbones favour sparse-event recipes (V7); logistic-normal backbones favour large-vocabulary recipes (V3).

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Preprint, 2026. Third manuscript in the CAOS_LDA_HSI programme. Companion papers: *Beyond Accuracy* (P1, twelve-axis framework) and *Which Wordification Matters?* (P3, 13-recipe sweep under fixed LDA backbone). Code and derived artefacts at https://github.com/fsantibanezleal/CAOS_LDA_HSI.

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3) A per-recipe *affinity score* that measures how a recipe transfers across the four backbones, identifying V3 (consistently top-3 across backbones) and V13 (consistently bottom-3) as the extremes.
4) A revised practical recommendation: the (backbone, wordification) pair is a single hyper-decision and should be reported jointly.

II. The four backbones

A. Online-VB LDA (P1, P3)

The canonical fixed-K LDA with online variational Bayes inference [?]. Symmetric Dirichlet priors $\alpha = 0.45$ on topic mixtures and $\eta = 0.20$ on topic-word distributions. K chosen by the same policy as P3: $K = \mathrm{clip}(\lfloor\overline{\mathrm{doclen}}/2\rfloor, 3, K_{P1})$ where $K_{P1} = \max(4, \min(12, n_{\mathrm{classes}}))$. Implementation: `sklearn.LatentDirichletAllocation`, random seed 42.

B. Hierarchical Dirichlet Process (HDP)

Bayesian nonparametric topic model with two-level stick breaking [?]. We use gensim's `HdpModel` with truncation $T = 20$ at the corpus level and $K = 8$ at the document level. Random seed 42. The number of effective topics is data-driven; the topic *rank* (the strength of each topic's prior weight) is part of the inference output.

C. Product LDA (ProdLDA)

Logistic-normal variational topic model [?]. Standard normal prior on the latent $\mathbf{z} \in \mathbb{R}^K$; $\theta = \mathrm{softmax}(\mathbf{z})$; two-layer MLP encoder, Pyro implementation. 80 epochs, batch 128, learning rate $1\mathrm{e}{-3}$, random seed 42.

D. Embedded Topic Model (ETM)

Joint word–topic embedding model [?]. Word embedding dimension 64; topic embeddings parameterised so the topic–word distribution is $\mathrm{softmax}(\beta_k^\top \rho)$ for word embedding matrix ρ. Same 80-epoch / batch-128 / seed-42 protocol.

III. Method: factorial design

The factorial design is fully crossed: for each pair (backbone, recipe) we use *the same* pre-computed doc-term matrix as in P3 (so the wordification stage is identical across backbones) and refit the backbone from scratch. For each cell we report:

- F-2 c_v coherence on the top-10 words per topic, using the same KSG-style sliding-window-NPMI estimator as P1.
- F-14 mean off-diagonal top-10 jaccard between topics (lower = more diverse).

The other axes (F-1 classification, F-7 NMI, F-22 counterfactual, F-18 reliability) are run for the LDA backbone only (carried over from P3) because they would require running a separate downstream classifier pipeline per backbone, which is deferred.

IV. Results

A. F-2 c_v factorial

Table II reports the mean F-2 c_v across the six labelled scenes for every (backbone, recipe) pair.

LDA's numbers in the V1–V12 columns are reproduced from P3; V13–V20 are new to this revision (c397–c401). HDP V13 cell is omitted because the recipe was not run under that backbone in the 2026-05-28 sweep. V20 (mutual-information-weighted bands) ties V3 / V12 for the LDA "best" slot and places in the top-3 of ProdLDA and ETM — confirming its versatility across non-HDP backbones.

B. Per-recipe affinity score

We define a recipe's *cross-backbone affinity score* $\mathcal{A}(\mathrm{V}_i) =$ the average over backbones of $(c_v(\mathrm{V}_i) - \min_j c_v(\mathrm{V}_j))/(\max_j c_v(\mathrm{V}_j) - \min_j c_v(\mathrm{V}_j))$ (math-mode). $\mathcal{A} \in [0, 1]$; higher means the recipe consistently scores near the per-backbone top across all four backbones.

V3 has the highest cross-backbone affinity: even though it only wins outright under ProdLDA, it is in the top-3 under every backbone, making it the most consistent recipe across families. **V13 (learned VQ-VAE) has the lowest affinity**, scoring near the per-backbone minimum across all four. The optimised reconstruction objective of VQ-VAE training does not align with any of the four backbones' priors.

a) V14, V18, V20 in the affinity ranking (2026-05-28).: Treating the new recipes' cells in Table II on the same affinity scale (with HDP V13 imputed at the per-recipe HDP average), V20 places second after V3 ($\mathcal{A} \approx 0.78$), slightly above V12. V14 lands around V1 ($\mathcal{A} \approx 0.42$), and V18 around V6 (≈ 0.39). The headline: **V20 is the most cross-backbone-portable label-aware recipe in the sweep**, and a near-replacement for V3 / V12 when label information is available.

C. Three backbone–recipe affinities

Dirichlet prior + dense vocabulary. LDA and ETM both use a Dirichlet-like prior over the topic simplex. They both pick V12 (GMM- token, 1600 word types from $B \times Q$ joint vocab with $B = 200, Q = 8$). The Dirichlet prior favours topics whose word distributions have meaningful mass on many tokens; V12's dense vocabulary delivers.

Truncation prior + sparse event vocabulary. HDP's stick-breaking truncation prior gives high weight to a few sticks (topics with concentrated mass) and decays geometrically thereafter. This rewards recipes whose tokens carry semantic weight individually rather than via density. V7 (absorption triplets, at most six tokens per document) fits this profile exactly.

Logistic-normal prior + large vocabulary. ProdLDA's logistic-normal prior over $\mathbf{z}$ allows the encoder to push topic mixtures arbitrarily far from the simplex centre. This rewards recipes whose vocabulary is large enough that a few tokens discriminate strongly between topics. V3's joint (band, bin) vocab ($B \times Q = 1600$ types) is the largest discrete vocabulary we consider with no spatial or learned structure — and it wins ProdLDA.

TABLE II
F-2 c_v MEAN ACROSS THE SIX LABELLED SCENES, 16×4 FACTORIAL (V1–V13 + V14, V18, V20 EXTENSION).

Backbone	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V18	V20	best
LDA	0.81	0.43	0.88	0.46	0.45	0.45	0.32	0.36	0.71	0.30	0.21	0.85	0.25	0.63	0.57	**0.85**	V3/V12/V20
HDP	0.54	0.44	0.31	0.44	0.44	0.32	**0.62**	0.31	0.35	0.35	0.50	0.34	—	0.31	0.42	0.38	V7
ProdLDA	0.75	0.44	**0.86**	0.43	0.43	0.54	0.51	0.31	0.73	0.33	0.34	0.83	0.25	0.47	0.48	0.74	V3
ETM	0.66	0.44	0.79	0.43	0.44	0.55	0.26	0.31	0.73	0.32	0.24	**0.82**	0.25	0.62	0.58	0.77	V12

TABLE III
CROSS-BACKBONE AFFINITY SCORES.

Recipe	$\mathcal{A}$
V3	0.83
V12	0.79
V9	0.62
V1	0.61
V6	0.45
V7	0.40
V5	0.31
V4	0.30
V2	0.21
V8	0.16
V11	0.16
V10	0.14
V13	0.04

V. PRACTICAL RECOMMENDATION

The pair $(\text{backbone}, \text{wordification})$ is a *single* hyper-decision in LDA-on-HSI work, not two independent choices. Reporting V12 as the best recipe is incomplete without specifying that the result holds for LDA and ETM; reporting V3 as the best recipe is incomplete without specifying ProdLDA.

We make three recommendations:

1) Future LDA-on-HSI papers should report the $(\text{backbone}, \text{wordification})$ pair explicitly in table headers, not as separate hyperparameters.
2) Cross-backbone affinity, as defined above, is a more informative summary than per-backbone-winner counts: it identifies recipes that generalise across families (V3, V12, V9, V1) versus those that only win in narrow regimes (V7 for HDP, V13 nowhere).
3) V1 (band-frequency) remains a defensible *reproducibility* canonical because of P3's F-18 result, but should not be claimed as a universal best on coherence in any backbone.

VI. LIMITATIONS

- LDVAE-T (Giannetti & Qureshi 2025) was the fifth proposed backbone; no public code at the time of writing. When code becomes available, the LDVAE-T row should be added.
- F-1 (Bayesian classification) is only run under LDA. Running F-1 under HDP / ProdLDA / ETM requires extending the topic-routed classifier pipeline to each backbone, deferred.
- HDP's $K_{\text{effective}}$ extraction (currently truncation level) needs refinement to use a topic-prevalence threshold; affects F-16 model-selection adequacy.